Describing CIFAR and LLM training theoretically: geometry, curvature, and useful Fanion modes

Third research attempt for the Neon repository

21 October 2026

Abstract

The most promising hypothesis for the neural experiments concerns the curvature-adjusted correction, rather than a flat raw gradient spectrum. If many singular directions require comparable Newton corrections, Muon can address them together while Fanion-k drops part of the useful correction. A precise local model proves this for every rank and describes the hybrid families through their angle in the curvature metric. It can favor Muon even when a Frobenius curvature surrogate favors Neon. For the actual experiments, CIFAR additionally requires normalized-weight geometry, batch-dependent losses, and label smoothing; the LLM speedrun requires shape-dependent steps, mixed optimizers, tensor batching, finite spectral filters, and an evolving attention window. We give a finite-horizon descent theorem accommodating these features, a sharp smoothed-loss bound on classification error, and measurements that could test the hypothesis. The local model is proved and numerically checked; its neural-network assumptions have not been established by the saved norm logs.

Recommended description and experimental scope

Describe the experiments as *stochastic, coupled optimization over matrix blocks and auxiliary parameters*, at a finite computational budget. The neural objective is not a sum of independent matrix quadratics. Layer-wise LMOs specify an update rule; they do not imply a separable loss. For CIFAR the appropriate geometric idealization is a product of weight spheres, with further scale redundancies from BatchNorm. For the LLMs the actual hidden-weight updates are additive; normalization of activations does not impose a sphere constraint on every weight matrix.

The comparison being explained.

The manuscript's eight-epoch CIFAR table compares Muon, Neon, Fanion-5, their F variants, S-Muon, NSGD, and SignSGD. Its LLM speedrun results mainly establish that Muon, F-Muon, and S-Muon are close; they do not supply an all-rank Neon/Fanion LLM experiment. An explanation of the latter ranking would therefore be a prediction for a future experiment. Classification accuracy and held-out token cross-entropy are also different targets.

Source audit.

The following facts come from the current source and manuscript; the scripts contain edited alternatives, not immutable manifests of every published run.

1. `code/cifar_bench/airbench_muon.py` uses six trainable convolution matrices, flattened as output-channels by input-channels times kernel area. Their shapes are 64×216, 64×576, 256×576, and three 256×2304 matrices. BatchNorm follows each convolution, with fixed scale and learned bias; the first also has max pooling before BatchNorm. The matrix optimizer handles these filters. An SGD optimizer handles the head, BatchNorm biases, and the trainable whitening bias.

2. The CIFAR script uses batches of 2,000, eight epochs (200 updates), random reshuffling and augmentation, and summed cross-entropy with label smoothing 0.2. It linearly decays the matrix learning rate. The standalone Muon and F-Muon classes normalize a weight *before* subtracting the update. The target Frobenius radius is the square root of its output width. The polynomial orthogonalization uses three iterations.

3. The manuscript explicitly separates the weight-normalized CIFAR accuracy table from its later gradient-norm runs *without* weight normalization. Those logs cannot directly verify tangent-direction or curvature hypotheses for the table's training algorithm.

4. There is a provenance inconsistency to resolve before a definitive experimental attribution: the current CIFAR script has a commented Neon call with `norm_weight=False` beside the reported 69.8% accuracy, although the manuscript describes the table as normalized. This does not recover the original run configuration. The normalized model below follows the manuscript's declared setting; the unnormalized alternative must be kept separate until the run settings are identified.

5. `code/gpt_pretrain/fmuon_pretrain.py` partitions parameters into hidden matrix tensors and an Adam group containing embeddings, output head, and scalars. It scales the hidden step by $\max(1, m/n)^{1/2}$ and any parameter multiplier. F and S components may use a different scaling flag; weight decay uses the base learning rate. There are multiple consecutive assignments to `optimizer2`; the last one wins. They cannot all describe one executed run.

6. The NanoGPT code has RMS normalization, residual paths, learned mixing scalars, zero-initialized output projections, and a bounded sigmoid transformation of the final logits. It trains on packed document tokens with a scheduled attention window. The gradient is a sum over local tokens, reduced across devices. The matrix momentum warms from 0.85 to 0.95 over 300 updates. These features matter more than a hypothetical global strong-convexity assumption.

7. A QKV parameter has shape $3 \times m \times n$. The polynomial treats its three matrices separately, but the F component in this script uses the Frobenius norm of the whole tensor. A correct block model must preserve that coupling. NanoGPT uses five repetitions of one polynomial; the Medium script uses five *different* coefficient triples. They are distinct filters.

All new work is in `theory_attempts`. We do not edit the submission, reconstruct missing checkpoints, fit neural losses, or claim new training results.

An objective in prediction space

Write the parameters as $\theta = (W_1, \ldots, W_L, z)$, where z contains parameters updated by SGD or Adam. For CIFAR the training objective is

$$F_{\mathrm{C}}(\theta) = \mathbb{E}_{\mathcal{B},a}\left[\frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \ell_{q_i}(s_\theta(x_i;\mathcal{B},a))\right], \qquad q_i = 0.8e_{y_i} + 0.02\mathbf{1}. \tag{1}$$

Here s is the final logit vector and a denotes augmentation. BatchNorm makes a prediction depend on the other examples in the batch, so this is an expectation of *batch losses*, not automatically the average of fixed per-example functions. Frozen running statistics define the evaluation predictor separately. Random reshuffling also does not give the usual conditionally unbiased IID gradient at every step.

For a language model with attention context rule c, write

$$F_c(\theta) = \mathbb{E}_{\text{document token position}}\Big[-\log p_\theta(x_j \mid x_{<j}; c)\Big].$$

(2)

Use *mean per-token* cross-entropy for theoretical comparisons. A fixed batch-size factor in summed gradients cancels in ideal homogeneous matrix oracles, but must be restored when specifying noise variance, epsilon effects, auxiliary optimizers, or an actual parameter displacement. Document dependence prevents interpreting every token as an independent gradient sample. For the speedrun $c = c_t$ changes, so the natural training objective is $F_t = F_{c_t}$. One can analyze fixed-window phases, or retain the objective drift explicitly. A theorem for a fixed full-context objective would instead have to account for the shorter-context gradient bias.

For fixed samples and context, let J_t be the Jacobian of final logits, $r_t = \nabla_s \ell$, and $C_t = \nabla_s^2 \ell$. For cross-entropy, $C_t = \operatorname{diag}(p) - pp^\top \succcurlyeq 0$ per prediction. The Gauss–Newton form of the joint curvature is

$$G_t = J_t^\top r_t, \qquad K_t^{\mathrm{GN}} = J_t^\top C_t J_t, \qquad \langle V, K_t^{\mathrm{GN}} V\rangle = \langle J_t V, C_t J_t V\rangle.$$

(3)

Sample/token averaging is implicit. $J_t V$ includes *all* matrix and auxiliary updates and their interactions. For a nonlinear model the true Hessian additionally contains $\sum_j (r_t)_j \nabla_\theta^2 (s_\theta)_j$. A rigorous upper curvature bound must control that residual and curvature changes along the step. At nonsmooth pooling/ReLU boundaries a local upper-model inequality can replace Hessian notation.

The Kronecker form $\langle V_l, A_l V_l B_l\rangle$ is a useful layer approximation to a Fisher or Gauss–Newton block; see Martens and Grosse (2015). It is not automatically a bound on the full neural Hessian. Cross-layer terms, correlations in convolutional patches, and BatchNorm coupling require estimates or a stated domination assumption. In particular, declaring both factors isotropic discards precisely the curvature structure that may explain Muon's advantage.

CIFAR: the exact normalization geometry

For a matrix of output width m_l, set $R_l = \sqrt{m_l}$ and $N_l(W) = R_l W / \|W\|_{\mathrm{F}}$. The current normalized implementations perform

$$H_{l,t} = \text{momentum from gradients at } W_t, \qquad W_{l,t+1} = N_l(W_{l,t}) - \gamma_{l,t} D_{l,t}(H_{l,t}).$$

(4)

This is not literally post-update projection. Nevertheless its normalized states $Y_{l,t} = N_l(W_{l,t})$ satisfy an exact retraction formula.

> **Proposition 1** (Normalization before a step). *For $\|Y\|_{\mathrm{F}} = R$, decompose $D = D_{\mathrm{T}} + \alpha Y$, where $\alpha = \langle Y, D\rangle / R^2$ and $\langle Y, D_{\mathrm{T}}\rangle = 0$. If $1 - \gamma\alpha > 0$,*
>
> $$N(Y - \gamma D) = N(Y - \eta D_{\mathrm{T}}), \qquad \eta = \frac{\gamma}{1 - \gamma\alpha}.$$
>
> (5)
>
> *Thus (4), viewed on normalized states, is a tangent step with a radial-dependent effective learning rate followed by a sphere retraction.*

Proof. Factor $Y - \gamma D = (1 - \gamma\alpha)(Y - \eta D_{\mathrm{T}})$ and use the invariance of N under positive scalar multiplication. A sufficient condition for the denominator is $\gamma \|D\|_{\mathrm{F}} < R$. □

For zero BatchNorm epsilon, positive scaling of each of these convolution weights leaves training predictions unchanged: convolution and max pooling scale together and BatchNorm cancels that scale before GELU. Per-channel positive scalings give additional redundancies. At the sphere representative, the relevant gradient is

$$G_l^{\tan} = G_l - \frac{\langle G_l, Y_l \rangle}{R_l^2} Y_l, \qquad \langle G_l^{\tan}, D_l^{\tan} \rangle = \langle G_l^{\tan}, D_l \rangle. \tag{6}$$

For an exactly scale-invariant training loss G_l is already tangent. If $W_l = c_l Y_l$, then $\nabla_{Y_l} F(Y, z) = c_l \nabla_{W_l} F(W, z)$. The momentum still contains gradients from previous, differently scaled states; it must be retained as actually computed rather than identified with the current tangent gradient.

Real epsilon, finite precision, and frozen evaluation statistics prevent treating all these statements as exact properties of the implemented evaluation predictor. They justify a clear idealization, with a residual term if needed. Without scale invariance one must directly retain $F(N(W)) - F(W)$ in an ambient-space descent argument, or study the actual two-stage map. One cannot silently replace it by projected gradient descent.

A concrete rank-dependent scale.

For the ideal plain Fanion direction $\|D_k\|_{\mathrm{F}} = \sqrt{k}$ and all six convolutions have $r_l = m_l$. Before tangent projection its relative step is

$$\frac{\gamma \|D_k\|_{\mathrm{F}}}{R_l} = \gamma \sqrt{k/m_l}. \tag{7}$$

At the inherited learning rate 0.24, the scales are:

Output width	Neon	Fanion-5	Muon	Frob.-matched Neon lr
64	0.030	0.0671	0.240	1.92
256	0.015	0.0335	0.240	3.84

These are direction-norm scales, not measured angles: projection, radial components, and the polynomial filter can change them. They show why inherited learning rates do not establish the best attainable performance of each rank. A useful ablation separately tunes every rank, includes $\gamma_{l,k} = \gamma_{l,r}\sqrt{r_l/k}$ as one starting rule, and measures the actual tangent step. Matching Frobenius size still does not match curvature, spectral norm, or prediction change; those are distinct controls.

The CIFAR mixture rates partially compensate for the half-sized polar component: $0.4/2 = 0.20$ for F-Muon and $0.42/2 = 0.21$ for S-Muon, versus 0.24 for Muon. For $a = b$, the extra F component has Frobenius size $1/\sqrt{r_l}$ relative to the full polar component, or 6.25–12.5% here. For S-Muon with $\eta = 0.003$, a dense sign component has ratio $\eta\sqrt{m_l n_l / r_l} = \eta\sqrt{n_l}$, or 4.4–14.4%. These small perturbations make similar behavior plausible after tuning. Curvature can amplify them, so Frobenius closeness alone is insufficient.

The earlier additive quadratic rank and nuclear-budget bounds do not transfer unchanged through normalization: scaling the current full weight adds a radial full-rank displacement. On normalized states that direction is a scale redundancy, not an ordinary rank-k correction to a fixed target.

LLMs: coupled blocks and finite spectral filters

Using a common schedule factor γ_t, represent the actual additive step as

$$\theta_{t+1} = \theta_t - \gamma_t V_t, \qquad (V_t)_l = \omega_l a_l \tilde{D}_{k_l}(H_{l,t}) + \tilde{\omega}_l b_l Z_{l,t} + \lambda_l W_{l,t}. \tag{8}$$

Here $Z = H/\|H\|_{\mathrm{F}}$ for F, and $Z = \eta \mathrm{sign} H$ for S; scaling can instead be absorbed into Z. ω_l includes shape/parameter multipliers; $\tilde{\omega}_l$ records whether the dense part is scaled. The auxiliary component of V_t is the actual Adam displacement divided by γ_t. Decoupled decay belongs in V_t; for a general matrix oracle it is not obtained by simply replacing G by $G + \lambda W$ inside the LMO.

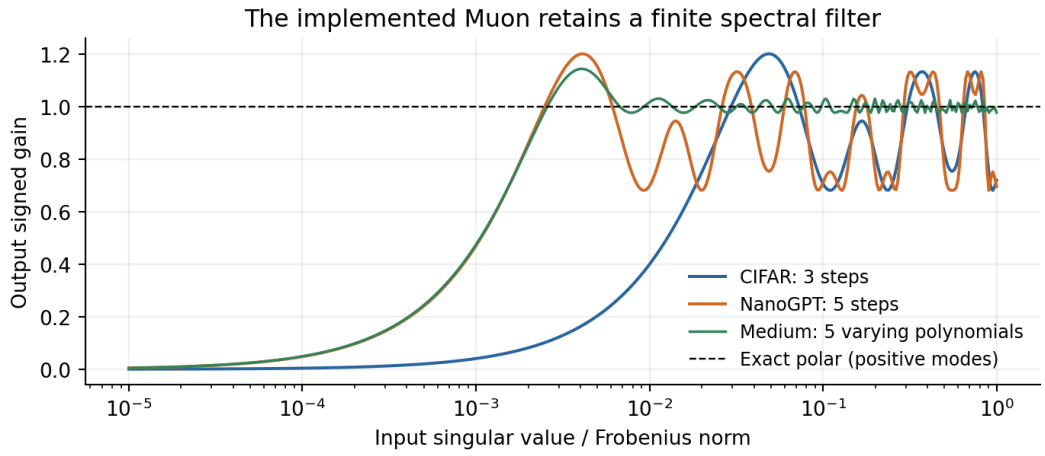
For a QKV tuple $H = (H_1, H_2, H_3)$, the NanoGPT plain spectral support is a sum across slices, while the F component is $(H_1, H_2, H_3)/(\sum_j \|H_j\|_{\mathrm{F}}^2)^{1/2}$. This differs from three separately unit-normalized dense components. Keep the code's normalization group as part of the theoretical block definition. The paper's textual sign coefficient and a script's `sign_lr_mult` should be translated into the *actual two component displacements*; they need not denote the same η in (8).

The implemented Muon is a spectral filter

Let $H = U\mathrm{diag}(s_i)V^{\mathsf{T}}$. Ignoring epsilon and rounding, a finite polynomial iteration gives

$$\tilde{D}_r(H) = U\mathrm{diag}(w_i)V^{\mathsf{T}}, \quad w_i = (p_q \circ \cdots \circ p_1)(s_i/\|H\|_{\mathrm{F}}), \quad p_j(x) = a_j x + b_j x^3 + c_j x^5. \tag{9}$$

This follows by substituting the SVD in each matrix iteration; it is also the scalar-filter description in Jordan's original writeup. For CIFAR $q = 3$ and NanoGPT $q = 5$, with repeated coefficients $(3.4445, -4.7750, 2.0315)$. Near zero their gains are respectively $40.87x$ and $484.88x$; zero modes remain zero. The Medium script instead uses five different polynomials, whose composition must be retained. Modes are neither uniformly set to one nor guaranteed to stay below one. Consequently an ideal feasible-polar LMO theorem needs an explicit approximation/feasibility correction, or can be replaced by the actual-direction theorem below.



Exact-arithmetic filters from the current scripts. Tiny modes remain attenuated after a finite number of iterations. This is a property of the update map, not evidence that attenuated modes are noise.

This supplies a useful middle model between hard truncation and an ideal polar update: preserve every direction with its actual gain w_i. It accommodates every Fanion rank, either with hard indicators or measured approximate gains, and the extra dense components. Whether weak modes carry signal or noise must be measured independently.

Momentum, dependence, and gradient alignment

The CIFAR buffer is $B_t = \beta B_{t-1} + g_t$, with $H_t = g_t + \beta B_t$. The LLM buffer is $M_t = \beta M_{t-1} + (1-\beta) g_t$, with $H_t = (1-\beta) g_t + \beta M_t$. For *constant* β and zero initialization, $M_t = (1-\beta) B_t$, so the two oracle inputs differ only by the positive factor $1-\beta$. Ideal homogeneous update maps coincide. This equivalence need not hold when momentum varies.

For the LLM's deterministic warmup schedule, the normalized input has weights

$$H_t = \sum_{s=0}^{t} w_{t,s} g_s, \quad w_{t,t} = 1 - \beta_t^2, \quad w_{t,s} = \beta_t (1-\beta_s) \prod_{j=s+1}^{t} \beta_j \ (s < t), \quad \sum_s w_{t,s} = 1 - \beta_t \prod_{j=0}^{t} \beta_j. \tag{10}$$

For constant momentum these reduce to the weights analyzed in `fanion_followup.pdf`. If $g_s = G_s + b_s + \xi_s$, with deterministic weights and martingale differences ξ_s of second moment at most σ_s^2, then

$$\mathbb{E}\left\|H_t - G_t\right\|_{\mathrm{F}} \le m_t \mathbb{E}\left\|G_t\right\|_{\mathrm{F}} + \sum_s w_{t,s} \mathbb{E}\left\|G_s - G_t\right\|_{\mathrm{F}} + \sum_s w_{t,s} \mathbb{E}\left\|b_s\right\|_{\mathrm{F}} + \left(\sum_s w_{t,s}^2 \sigma_s^2 \right)^{1/2}, \quad m_t = 1 - \sum_s w_{t,s}. \tag{11}$$

The proof is the triangle inequality and orthogonality of martingale increments. Changes in parameters *and context rule* enter $G_s - G_t$. Bias b_s allows reshuffling, document-stream dependence, or an objective mismatch. This is a conditional noise description, not a claim that token gradients are IID or that variance divides by token count.

Most importantly, the useful alignment is $\langle \nabla F_t, D_k(H_t) \rangle$, not the always-positive self-alignment $\langle H_t, D_k(H_t) \rangle$. Finite-batch nonlinear transformations may be biased. Do, Dereich, and Jentzen (2026) give relevant stochastic counterexamples and batch-size-dependent error analysis for generalized Muon. This does not show failure on these neural tasks; it rules out treating fixed-batch consistency as automatic.

A finite-horizon theorem for both settings

Let x_t be the additive LLM state, or the sphere representatives and auxiliary parameters for the scale-invariant CIFAR idealization. Write the step as $x_{t+1} = \mathcal{R}_{x_t}(-\gamma_t V_t)$, using the identity map for additive updates and a product retraction for CIFAR. In the latter case the lth component of V_t is $(\eta_{l,t}/\gamma_t) D_{l,t}^{\tan}$ from (5). Retain auxiliary updates in the same vector.

> **Theorem 2** (Actual-direction descent with objective drift). *Suppose along the executed steps that*
>
> $$F_t(\mathcal{R}_{x_t}(-\gamma_t V_t)) \le F_t(x_t) - \gamma_t A_t + \frac{1}{2}\gamma_t^2 B_t, \qquad A_t = \langle \mathrm{grad} F_t(x_t), V_t \rangle, \quad B_t \ge 0. \tag{12}$$
>
> *Define* $\delta_t = F_{t+1}(x_{t+1}) - F_t(x_{t+1})$ *and assume* $F_T(x_T) \ge F_{\min}$. *For every realized trajectory,*
>
> $$\sum_{t<T} \gamma_t A_t \le F_0(x_0) - F_{\min} + \frac{1}{2} \sum_{t<T} \gamma_t^2 B_t + \sum_{t<T} \delta_t. \tag{13}$$
>
> *Replacing the last sum by* $\sum_t [\delta_t]_+$ *gives a conservative bound. Taking expectations preserves the inequality when the terms are integrable.*

Proof. Add δ_t to (12) and sum; the objective values telescope. No independence or convexity is used. For a fixed context all δ_t vanish. An additive upper-model error can be added to the right-hand side and telescopes in the same way. □

The theorem exposes the quantities a rate comparison must retain: alignment with the target gradient, curvature of the *actual joint step*, and context/objective drift. Substituting a global Frobenius smoothness bound is valid when available but can destroy the comparison. For example, a structured certificate could bound B_t by $\chi_t[\sum_l \langle V_l, A_l V_l B_l\rangle + C_z(V_z)]$ after proving a cross-block and nonlinear-residual bound. A raw Hessian-vector product at one point is an estimate, not itself a certificate over a finite segment.

Optional conditional linear rate.

In a fixed-objective phase, suppose $A_t > 0$, $B_t > 0$ and a local Polyak–Łojasiewicz condition holds above the target loss: $\|\mathrm{grad}F\|_{\mathrm{F}}^2 \geq 2\mu(F - F_*)$. Define the direction quality $q_t = A_t^2/(B_t\|\mathrm{grad}F\|_{\mathrm{F}}^2)$ and schedule ratio $\rho_t = \gamma_t B_t/A_t$. If $\rho_t \in (0, 2)$, (12) gives

$$F(x_{t+1}) - F_* \leq [1 - \mu q_t \rho_t(2 - \rho_t)](F(x_t) - F_*). \tag{14}$$

This follows by writing the decrease as $A_t^2\rho_t(2 - \rho_t)/(2B_t)$. It identifies both a direction factor and a tuning factor. Neither a global PL condition nor uniformly positive alignment has been established for these training runs. In a noisy regime use an expected version with explicit alignment/error assumptions; do not infer this deterministic rate from same-batch norms.

A more useful hypothesis: comparable Newton corrections

The key question is which modes need changing in prediction space after curvature is accounted for. A small gradient singular value can reflect small curvature, rather than a completed or useless mode. This suggests testing *broad Newton corrections* instead of assuming a broad raw gradient spectrum.

At a fixed point let $P_i = u_i v_i^\top$ be the singular directions of the actual oracle input. For CIFAR replace these by their tangent projections and use the retraction pullback. Define

$$g_i = \langle \mathrm{grad}F, P_i\rangle, \qquad K_{ij} = \langle P_i, \mathcal{H}P_j\rangle, \qquad Q(d) = \frac{(g^\top d)^2}{d^\top K d} \quad (g^\top d > 0). \tag{15}$$

Here $\mathcal{H}$ is an exact positive quadratic curvature or a valid local upper model. For an exact quadratic, $Q(d)/2$ is the loss decrease under the best scalar step in direction $\sum_i d_i P_i$. The rank-k plain direction has $d_i = \mathbf{1}_{i \leq k}$. Finite Muon uses the gains in (9); F adds $bs_i/\|H\|_{\mathrm{F}}$ to its spectral weights. For S, retain sign$H$ as an extra direction and extend g, K with its curvature and cross terms. Its component outside the span of the P_i must not be dropped.

> **Proposition 3** (Newton-aligned full polar direction). *Suppose $K \succ 0$ on these coordinates and*
>
> $$g = eK\mathbf{1}, \qquad e > 0. \tag{16}$$
>
> *Then the full ideal Muon direction is the optimal direction in this local quadratic model, and for every Fanion or hybrid coefficient vector d,*

$$\frac{Q(d)}{Q(\mathbf{1})} = \frac{(\mathbf{1}^\top K d)^2}{(\mathbf{1}^\top K \mathbf{1})(d^\top K d)} = \cos_{K2}(\mathbf{1}, d) \le 1. \tag{17}$$

Only directions of positive alignment are used. If the actual Newton correction has zero coordinates outside the polar span, the same statement holds after adding other directions, including the sign direction.

Proof. The minimizer of the quadratic model has correction $K^{-1}g = e\mathbf{1}$. Substitution gives (17); its upper bound is Cauchy–Schwarz in the K inner product. The optimal scalar step for the full polar direction is e. □

This is a sufficient, testable structural hypothesis, not a conclusion deduced from the choice of optimizer. It can also be used with the *implemented* full-filter direction w: the corresponding hypothesis is $g = eKw$. The local theorem is then exact for that direction. Replacing an indefinite Hessian by a PSD Gauss–Newton or trust-region upper model changes the hypothesis and must be stated.

Robustness of the hypothesis.

If $g = eK\mathbf{1} + h$ and $\left\|K^{-1/2}h\right\|_{\mathrm{F}} \le \delta e \left\|K^{1/2}\mathbf{1}\right\|_{\mathrm{F}}$, $0 \le \delta < 1$, then the full polar score relative to the best possible local score obeys

$$\frac{Q(\mathbf{1})}{g^\top K^{-1} g} \ge \left(\frac{1-\delta}{1+\delta}\right)^2. \tag{18}$$

Indeed its numerator's alignment is at least $e(1-\delta)\mathbf{1}^\top K\mathbf{1}$, while $\left\|K^{-1/2}g\right\|_{\mathrm{F}} \le e(1+\delta)\left\|K^{1/2}\mathbf{1}\right\|_{\mathrm{F}}$. Likewise, under exact (16), a hybrid $d = a\mathbf{1} + bz$ with $\left\|K^{1/2}bz\right\|_{\mathrm{F}} \le \varepsilon a \left\|K^{1/2}\mathbf{1}\right\|_{\mathrm{F}}$, $a > 0$, $\varepsilon < 1$, retains at least $((1-\varepsilon)/(1+\varepsilon))^2$ of the full score. This allows a negative b as in the manuscript's $\alpha > 1$ experiments, provided the perturbation bound holds; a convex-LMO-sum argument would not cover that case.

Every Fanion rank in a solvable curvature model

Suppose the modal curvature is diagonal with entries $\lambda_i > 0$ and the current modal errors are $e_i > 0$. Then $g_i = \lambda_i e_i$. When $e_i = e$ for all i, for the set I_k selected from the oracle input,

$$Q_k = e^2 \sum_{i \in I_k} \lambda_i, \qquad \frac{Q_r}{Q_k} = \frac{\sum_{i \le r} \lambda_i}{\sum_{i \in I_k} \lambda_i}. \tag{19}$$

For an exact-gradient input $H = G$, I_k comprises the k largest λ_i. For a general momentum input the denominator is at most that largest-k curvature mass, giving a lower bound on the score ratio. The effective rank governing the full-versus-top-mode comparison is the *curvature mass* $\sum_i \lambda_i/\lambda_1$ in the exact-gradient case, rather than $(\sum_i \lambda_i)^2/(r\lambda_1^2)$ from a Frobenius surrogate. For $0 < e_{\min} \le e_i \le e_{\max}$, even if sorting by gradient and sorting by curvature differ,

$$\frac{Q_r}{Q_k} \ge \left(\frac{e_{\min}}{e_{\max}}\right)^2 \frac{\sum_i \lambda_i}{\sum_{i \le k} \lambda_i^{\downarrow}}. \tag{20}$$

For the numerator use $(\sum_i \lambda_i e_i)^2 / \sum_i \lambda_i \ge e_{\min 2} \sum_i \lambda_i$; for the selected denominator use $Q_k \le e_{\max 2} \sum_{i \in I_k} \lambda_i \le e_{\max 2} \sum_{i \le k} \lambda_i^{\downarrow}$.

For any positive-alignment spectral filter or hybrid weights d_i and equal errors, the fraction of the optimal decrease is

$$\frac{\mathcal{Q}(d)}{e^2 \sum_i \lambda_i} = \frac{(\mathbb{E}_\pi d)^2}{\mathbb{E}_\pi d^2} = \frac{1}{1 + \mathrm{Var}_\pi(d)/(\mathbb{E}_\pi d)^2}, \qquad \pi_i = \frac{\lambda_i}{\sum_j \lambda_j}. \tag{21}$$

Thus Muon's near-constant weights can be good in the curvature metric, F-Muon's weights can be a modest perturbation, and truncated Fanions lose useful curvature mass. A sign update may have extra off-modal curvature; its behavior additionally depends on the entry basis. These are explicit mechanisms for closeness of the full-rank hybrids and weakness of small ranks, with no universal ordering when the structural hypothesis fails.

An exactly embedded matrix example.

Take $r = 128$, $\lambda_i = i^{-0.9}$, any orthogonal U, V, and

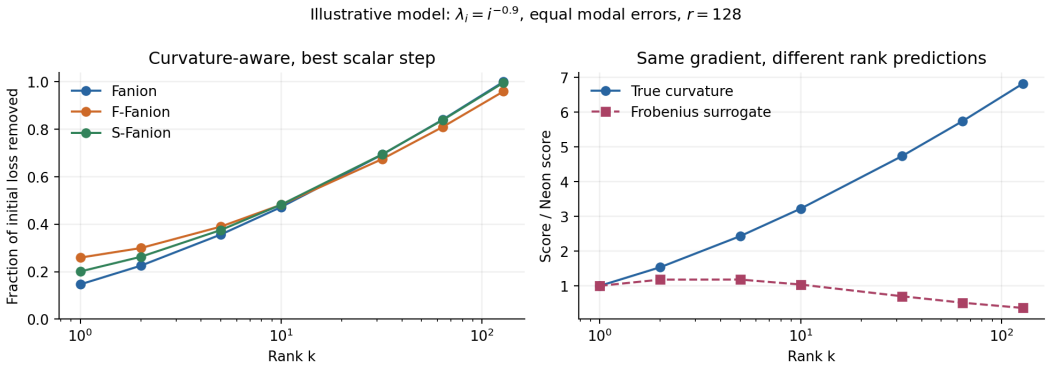
$$f(X) = \tfrac{1}{2}\langle X, AXB\rangle, \qquad A = U\mathrm{diag}(\sqrt{\lambda_i})U^\top, \quad B = V\mathrm{diag}(\sqrt{\lambda_i})V^\top, \quad X_0 = UV^\top. \tag{22}$$

The gradient is $U\mathrm{diag}(\lambda_i)V^\top$ and $AP_iB = \lambda_i P_i$, so the example realizes the modal model exactly. True curvature favors Muon over Neon by 6.82127, whereas the common Frobenius surrogate gives ratio 0.363514 and favors Neon. This shows why the latter surrogate can give an incorrect ranking without any noise.

For a reproducible random rotation, $a = b = 1/2$, and sign scale 0.003, the fractions of initial loss removed by the best scalar step are:

Rank	Plain	F	S
1	0.1466	0.2594	0.2009
5	0.3562	0.3897	0.3753
128	1.0000	0.9595	0.9961

For the three current full-rank polynomial filters the corresponding fractions are 0.9360 (CIFAR coefficients, three steps), 0.9714 (NanoGPT, five steps), and 0.99973 (Medium's varying polynomials). These are illustrative exact-arithmetic calculations, not measured network curvatures, training times, or a fitted explanation of every empirical rank comparison. Finite-step overshoot, different trajectories, and stochastic alignment can change the ordering. A local optimum over scalar steps is also different from the actual fixed schedule.



The same exactly specified matrix model across Fanion ranks. Left: fraction of loss removed after the best scalar step, including both hybrids. Right: retaining directional curvature reverses the rank trend suggested by a Frobenius surrogate. Neither panel uses neural training data.

What convergence target matches CIFAR accuracy?

For C classes and smoothing $\varepsilon \in [0, 1)$, let $q_h = 1 - \varepsilon + \varepsilon/C$ and $q_l = \varepsilon/C$. For an example with correct label y,

$$\ell_q(p) = H(q) + \mathrm{KL}(q\|p), \qquad \kappa_{\varepsilon,C} = q_h \log\frac{2q_h}{q_h + q_l} + q_l \log\frac{2q_l}{q_h + q_l}. \tag{23}$$

Use the convention $0\log 0 = 0$.

> **Proposition 4** (A sharp mistake cost for smoothed cross-entropy). *If a class other than y has probability at least p_y, then $\ell_q(p) - H(q) \geq \kappa_{\varepsilon,C}$. Consequently, for the same prediction mechanism and distribution,*
>
> $$\Pr(\widehat{y} \neq y) \leq \frac{\mathbb{E}\ell_q(p) - H(q)}{\kappa_{\varepsilon,C}}. \tag{24}$$

Proof. For a fixed competing class j, minimize $\mathrm{KL}(q\|p)$ subject to $p_j \geq p_y$. The optimum pools the two coordinates: $p_y = p_j = (q_h + q_l)/2$, with $p_i = q_i$ otherwise. This follows from the Lagrange conditions, or convexity of $-\sum_i q_i \log p_i$. The resulting KL is (23), independent of j. Take the union over possible wrong maximizers pointwise (each has the same minimum cost), then average the indicator inequality. □

For the actual CIFAR smoothing $\varepsilon = 0.2$, $C = 10$,

$$H(q) = 0.86689391, \qquad \kappa_{0.2,10} = 0.48773025, \qquad \log(q_h/q_l) = \log 41 = 3.71357207. \tag{25}$$

Units are natural-log cross-entropy. Perfect training classification does not require vanishing gradients or attaining $H(q)$: it only requires the correct class to have the largest probability. Moreover the smoothed logit optimum has a finite target margin $\log 41$; a zero-loss or unbounded-margin model for unsmoothed separable classification would misdescribe this experiment.

The inequality links training optimization to training error, or population loss to population error. For validation accuracy, one additionally needs the appropriate loss generalization control and a bound on changing the prediction mechanism (augmentation, frozen BatchNorm statistics, and test augmentation). A training-mode gradient theorem alone does not rank the reported validation accuracies. Even (24) is an upper bound, so comparing its values cannot prove a strict actual-accuracy ordering.

For LLM evaluation, the target is already held-out mean token cross-entropy. There is no need to replace it by classification accuracy or by a vanishing parameter-gradient target. Population versus finite held-out loss still requires a sampling statement if a statistical claim is intended. The bounded logit transformation in the NanoGPT script further means that the usual unrestricted softmax infinite-margin limit is not its literal model.

Measurements that would decide between explanations

The new structural assumption is not checkable from Frobenius, operator, and nuclear norms alone. The following measurements at a few shared checkpoints would make the theory discriminating.

1. **Record the actual configuration and displacement.** For every group save the normalization flag and order, matrix/tensor shape, learning-rate multipliers, the two hybrid component steps, polynomial coefficients, momentum convention, and auxiliary update. For CIFAR also save W, $N(W)$, the radial component, and the tangent displacement. For the LLM save context/window and summed-versus-mean gradient scaling.
2. **Use an independent probe gradient.** At the same state and context, form the candidate directions from the same actual momentum input H, then estimate $\langle \nabla F_t, V \rangle$ on fresh training batches/documents. This diagnoses the update map without its same-batch self-alignment bias. BatchNorm state must be kept consistent during probes. At each algorithm's own trajectory also log its own H; a common-input counterfactual does not replace that.
3. **Measure curvature and actual loss change.** Estimate $V^{\mathrm{T}} K^{\mathrm{GN}} V$ by a joint Jacobian-vector product and compare with a Hessian-vector product or symmetric small loss probes. Include matrix/auxiliary and cross-layer interactions, and for CIFAR probe the actual retraction. Check finite-step Taylor error rather than treating a pointwise curvature as an upper bound. Both A^2/B and the actual scheduled decrease $\gamma A - \gamma^2 B/2$ matter.
4. **Resolve all ranks.** Log leading singular values of H beyond ranks 1 and 5, the polynomial gains, and $g_i = \langle G_{\mathrm{probe}}, P_i \rangle$. Estimate the modal Gram matrix or selected diagonal/cross terms. The question is whether the inferred correction $K^{-1} g$ is broad and approximately constant in the polar modes, and whether curvature mass outside the first k directions is large. Do not infer this from a heavy-tailed gradient spectrum alone.
5. **Separate tuning and generalization.** Compare inherited, independently tuned, and tangent/Frobenius-matched rank schedules. Keep normalized and unnormalized CIFAR runs separate. For an accuracy explanation, examine train loss, validation loss, and the normalization/evaluation gap alongside accuracy. For LLMs compare fixed context phases or report the context drift.

These measurements yield competing predictions. If the useful correction is broad but gradients are anisotropic, the curvature model predicts a Muon advantage and near-equivalence of small full-rank perturbations. If weak modes are mostly noise, their independent alignments will be small or negative and a truncated/soft-filter method can be better. If the gap largely vanishes after rank-dependent retuning, step scale explained much of the original inherited-hyperparameter gap. The results should be used to select a structural hypothesis, not to assert that one already holds.

What is established, and what is still a hypothesis

Established here are the exact normalization reindexing, finite-filter representation, variable-momentum weights, drift-aware descent theorem, local curvature/Newton comparison for every rank, the smoothed-loss

classification bound, and an exactly embedded example reversing a Frobenius-based rank prediction. The analysis script passes 3,378 numerical checks and records input hashes and limitations in `results/deep_learning/summary.json`. Its examples are deterministic mathematical diagnostics, with random seed 20261021, not neural experiments.

For the article, the strongest defensible description is: *"Muon can exploit a broad useful correction in the network's local curvature geometry. Low-rank Fanions can miss a substantial fraction of that correction; full-rank F and S perturbations can retain most of it. The explanation depends on projected/functional alignment, directional curvature, and the effective schedule, which are not fixed by asymptotic exponents or by three raw gradient norms."* This remains a proposed mechanism for the CIFAR/LLM results until the stated quantities are measured. It does not establish a universal optimizer ordering, a global neural convergence rate, or a generalization theorem.

Reproduce.

From the repository root:

```
python theory_attempts/deep_learning_analysis.py
python theory_attempts/build_report.py --stem deep_learning_model
```

The CSV files contain the full rank curves, nominal CIFAR scales, and spectral filters; standalone figures are provided as PDF and PNG. `paper_insert_deep_learning.tex` is a compact candidate discussion and proposition for the submission, which remains unmodified.

References and relation to existing work

1. Martens and Grosse (2015), Optimizing Neural Networks with Kronecker-factored Approximate Curvature. Provides the established Fisher-block/Kronecker motivation. We use this as a possible model, not as a theorem certifying the full neural Hessian.
2. van Laarhoven (2017), L2 Regularization versus Batch and Weight Normalization. Studies scale invariance and effective learning rates under normalization. The normalization-before-step identity here is a direct calculation for this repository's order of operations.
3. Jordan (2024), Muon: An optimizer for hidden layers in neural networks. Describes polynomial action on singular values, mixed parameter groups, and the motivation for amplifying weaker directions. Our mode model gives a conditional curvature interpretation; it is not a claim that the spectral-filter representation is new.
4. Do, Dereich, and Jentzen (2026), On MUON optimization: From non-convergence to an error analysis with Polar Express and the Newton–Schulz polynomial from implementations. Gives generalized polynomial-Muon analysis and stochastic examples with a batch-size error term, reinforcing the need to retain noise-induced alignment error.
5. For prior structured convergence results and multiblock norm models, see Kovalev and Borodich (v2, 2026), Shen et al. (v3, 2026), and Crawshaw et al. (2025). These were discussed in the earlier attempts. The present note contributes an experiment-specific description and conditional mechanisms, not a priority or novelty claim for standard local descent and metric-angle facts.